Hyperspectral and Multispectral Image Fusion via Graph Laplacian-Guided Coupled Tensor Decomposition

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***Abstract*— We propose a novel graph Laplacian-guided coupled tensor decomposition (gLGCTD) model for fusion of hyperspectral image (HSI) and multispectral image (MSI) for spatial and spectral resolution enhancements. The coupled Tucker decomposition is employed to capture the global interdependencies across the different modes to fully exploit the intrinsic global spatial–spectral information. To preserve local characteristics, the complementary submanifold structures embedded in high-resolution (HR)-HSI are encoded by the graph Laplacian regularizations. The global spatial–spectral information captured by the coupled Tucker decomposition and the local submanifold structures are incorporated into a unified framework. The gLGCTD fusion framework is solved by a hybrid framework between the proximal alternating optimization (PAO) and the alternating direction method of multipliers (ADMM). Experimental results on both synthetic and real data sets demonstrate that the gLGCTD fusion method is superior to state-of-the-art fusion methods with a more accurate reconstruction of the HR-HSI.**

***Index Terms*— Coupled tensor decomposition, graph Laplacian, hyperspectral imaging, image fusion, manifold structure.**

I. Introduction

THE wealth of spectral characteristics acquired by hyperspectral imaging containing hundreds of contiguous bands can immensely benefit many image processing and remote sensing applications, such as object classification [1]–[3], object detection [4], [5], and denoising [6]–[8], to name a few. HSIs with the high spatial resolution are in great demand. However, due to the sensors' hardware limitation in optical imaging, there is always a tradeoff between spectral and spatial resolutions, and most satellite hyperspectral missions generally have a coarser spatial resolution.

Recently, fusion methods based on some generic priors have been investigated for spatial resolution enhancement with an HR-MSI and a low-resolution (LR)-HSI [9]–[12]. Advances in matrix factorization techniques have motivated significant research efforts to solve the ill-posed HSI-MSI fusion problem. These models decompose the unfolding of the HR-HSI into a coefficient matrix and a basis matrix under some priors. The pioneering work of matrix factorization [13] formulates the fusion problem as a spectral unmixing problem: the spectral basis (or end-member reflectance) is first estimated from the input LR-HSI followed by the learning of sparse coefficient matrix. To achieve reliable reconstructions, the works in [12], [14], [15] utilize a coupled matrix factorization scheme that alternately updates coefficient matrix and spectral basis matrix with nonnegative constraints. All the fusion models discussed earlier hypothesize that each pixel spectrum is a linear combination of the spectral signatures of pure end-members. However, these methods can be generalized as a matrix factorization directly being applied to the whole image with nonnegative or sparse constraints, rather than taking into account the local spatial information. The lack of introducing local spatial information tends to deteriorate the fusion performances, typically the deficiency of preserving spatial details. To alleviate this issue, state-of-the-art fusion methods [16]–[19], [20], [21] aim at fully exploiting the global and local spatial information of HR-HSI. A major disadvantage of conventional matrix-based fusion methods is the unfolding process of the third-order HSI into matrices, which breaks the HSI structures and prone to loss of structural information. It becomes hard to exploit the spectral correlations of the HR-HSI and results in distortions of spectral features.

In order to account simultaneously for all spectral–spatial information associated with the HR-HSI, tensor decomposition has been commonly utilized with a strength simultaneously capturing the correlations among different modes. In a Candecomp/Parafac (CP) decomposition framework, the tensor data can be represented by the smallest number of rank-1 tensor components. Kanatsoulis *et al.* [22] proposed a coupled CP

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decomposition framework to tackle the fusion of HSI and MSI. Though the correlations among different dimensions are captured to some degree, the local spatial information of HR-HSI has not been taken into consideration yet. More precisely, in terms of the CP decomposition itself, the introduction of some spatial image priors to CP decomposition is likely to make the essential uniqueness of CP decomposition not hold. On the contrary, the Tucker decomposition framework tackles partially with the drawbacks of CP decomposition due to its flexibility and generality. The Tucker tensor decomposition is a reliable model that not only revealing the global structural information of a tensor object based on strong interactions among each mode but also allowing the introduction of some complementary information of the HSI. Recently, the HSI-MSI fusion problem has been formulated from the viewpoint of the Tucker decomposition [23]–[26]. A graph-regularized low-rank tensor representation is proposed in [23] to infer the spectral smoothness and spatial consistency. Due to the extremely large Laplacian matrix, the method described in [23] is so time-consuming that one cannot effectively conduct in practice. To further improve fusion performance, the idea of the coupled sparse Tucker decomposition model is used to promote the sparsity of core tensor and to estimate the dictionaries of the row, column, and spectral modes seriatim from a global perspective [26]. The global sparsity representation performed on core tensor inherently builds a bridge among different modes. However, the sparse core tensor and dictionaries obtained from the observed LR-HSI and HR-MSI may be biased from the original ones of the reference HR-HSI because the Tucker decomposition has been proven to preserve global statistics but may interpret local structural information inadequately in the multilinear space. More recently, the HSI-MSI fusion problem in [24] and [25] has been formulated as a sparse Tucker decomposition scheme. Each 3-D cube is approximated by learning the spectral dictionary from the LR-his, and the two spatial dictionaries of the row and column modes from the HR-MSI. Nonlocal similarities are incorporated by hypothesizing that similar 3-D cubes of the HR-HSI should share the same dictionaries in the sparse tensor factorization scheme. However, the methods abovementioned argue that the spectral–spatial information is embodied in the original image domain, which merely reflects the intuitional structure of HR-HSI. Suppose that data samples of high-dimensional data are drawn from data manifold, and then, the local similarities of the data samples can be regarded to reside on local submanifolds. Therefore, the underlying local spectral–spatial information can be further recovered in the manifold domain. Since the intrinsic local submanifolds of high-dimensional data are often hidden in low-dimensional embedding space, the graph embedding techniques can be deployed to learn such an embedding space while preserving the submanifolds of data [27]. Yet, extending the graph embedding techniques to the multilinear space is somewhat challenging because they fail to characterize the intrafactor and interfactor relations of high-dimensional data. The study in [28] indicates that local information embedded in the latent manifold domain with multilinear graph embedding techniques formulated by the graph Laplacian regularization generally performs better in recognition tasks because the local structure of data is particularly preserved. From this perspective, it still exists much room for improvements in modeling the high-dimensional data in multilinear space if the explicit structure or latent factors are well-formulated. Therefore, the graph Laplacian regularizations imposed on factor matrices in the multilinear space and Tucker decomposition can be integrated into a unified fusion framework for complementing to the deficiencies of the Tucker decomposition. The latent local information embedded in the manifold domain revealing implicitly more structural priors of spatial–spectral information has yet to be addressed so far in the task of HSI-MSI fusion. What even more interesting is to build a bridge between tensor decomposition and graph embedding scheme by simultaneously exploiting the global and local structures of HR-HSI.

To address the limitations of existing tensor-based methods by the strength of the multilinear graph embedding techniques, we propose a novel HSI-MSI fusion method that incorporates both the graph Laplacian regularization and the coupled Tucker decomposition (gLGCTD) into a unified framework. The proposed method characterizes the global spatial–spectral information captured by the coupled Tucker decomposition and latent local submanifolds of the HSIs. To be exact, in order to exploit the high-dimensional characteristics of HSIs and locally preserve the underlying spatial–spectral information, we incorporate the complementary submanifold structures embedded in HR-HSIs into the coupled Tucker decomposition framework. The submanifold structures encoded by the graph Laplacian regularization can be disentangled with two spatial and spectral factor graphs. In this way, the disadvantages of the multilinear-based Tucker decomposition are significantly rectified via leveraging multilinear graph embedding techniques formulated by the graph Laplacian regularizations imposed on factor matrices. To the best of our knowledge, this article represents the first effort to incorporate the underlying local submanifold information of spatial–spectral structures in HSI and global spatial–spectral structures from the Tucker decomposition into a unified framework. It should be noted that the coupled Tucker decomposition framework in our proposed model is different from the formulations in [26] as their models do not fully exploit the local geometry and the underlying manifold structures of HSIs. Finally, the gLGCTD fusion framework can be solved by a hybrid framework between PAO and ADMM to produce more accurate reconstructions. Experimental results on several synthetic and real data sets demonstrate that the proposed gLGCTD fusion method is superior to state-of-the-art fusion methods with respect to visual comparison and numerical analysis.

The main contributions to this article are summarized as follows.

1) We propose a unified global and local fusion framework for the task of HSI and MSI fusion that not only preserves global spatial–spectral information but also adequately interprets local geometry of HSI.
2) The proposed gLGCTD method is a tensor-based method that preserves the intrinsic spatial–spectral

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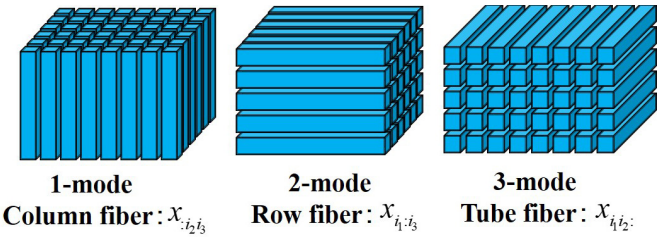


Fig. 1. Three modes of a third-order tensor.

information and exploits the latent submanifold structures encoded by the graph Laplacian regularization.

3) We propose an efficient algorithm via a hybrid framework between PAO and ADMM to formulate the fusion problem and compare it with state-of-the-art fusion methods on several synthetic and real data sets.

The remainder of this article is organized as follows. In Section II, preliminaries on tensor basics are detailed. In Section III, we describe the gLGCTD HSI-MSI fusion framework and the effective optimization procedure. Section IV presents the experimental results and discussions, and Section V makes overall conclusions.

II. Preliminaries of Tensor Basics

For convenience, some necessary definitions and preliminaries of tensors are introduced first. Scalars, vectors, and matrices are denoted by lowercase italics letters, bold face lowercase letters, and boldface capital letters, respectively. A third-order tensor is denoted by the Euler script letter $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$. The element in $\mathcal{X}$ can be denoted as $\mathbf{x}_{i_1,i_2,i_3}$. $\|\mathcal{X}\|_F = (\sum_{i_1,i_2,i_3} |x_{i_1,i_2,i_3}|^2)^{1/2}$ denotes the Frobenius norm [29] of tensor $\mathcal{X}$. Tensors can be regarded as the higher order generalizations of matrices and vectors [30]. A third-order tensor $\mathcal{X} = (X_1, X_2, \ldots, X_{i_3}) \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ consists of the collection of matrices $X_{i_3} \in \mathbb{R}^{I_1 \times I_2}$.

In tensor operations, fibers are the high-order analog of matrix rows and columns, which can be defined by fixing every index but one [29]. For example, a matrix column is a 1-mode fiber, and a row is a 2-mode fiber. As shown in Fig. 1, a third-order tensor has 1-mode column, 2-mode row, and 3-mode tube fibers, denoted by $\mathcal{X}_{:i_2i_3}$, $\mathcal{X}_{i_1:i_3}$, and $\mathcal{X}_{i_1i_2:}$, respectively. The n-mode unfolding of a third-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ can be denoted as $X_{(n)}$, which is stacked by all the n-mode vectors as its columns [29]. The third-order tensor element (i_1, i_2, i_3) maps to matrix element (i_n, j), where $j = 1 + \sum_{k=1, k \neq n}^{3} (i_k - 1) J_k$, with $J_k = \prod_{m=1, m \neq n}^{k-1} I_m$. The n-mode product of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ with a matrix $A \in \mathbb{R}^{J \times I_n}$ is denoted by $\mathcal{Y} = \mathcal{X} \times_n A$, where $n = 1, 2, 3$. Elementwise, we have $(\mathcal{X} \times_n A)_{i_{n-2} i_{n-1} j i_{n+1} i_{n+2}} = \sum_{i_n=1}^{I_n} x_{i_1 i_2 i_3} a_{j i_n}$. Each n-mode fiber is multiplied by the matrix A, and the n-mode product can also be expressed as $Y_{(n)} = A X_{(n)}$ in terms of the n-mode unfolding operation. The Tucker decomposition reflects the primary properties of the original tensor. As a form of high-order PCA, it splits a tensor into a core tensor multiplied by orthogonal factor matrices. Through the Tucker decomposition, tensors can be effectively projected onto much lower subspaces [30]. Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ be a third-order tensor, and then, the Tucker decomposition can be expressed as

$$\mathcal{X} = \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 V_3 \tag{1}$$

where V_1, V_2, and V_3 can be regarded as the principal components of each individual dimension. The core tensor $\mathcal{C} \in \mathbb{R}^{J_1 \times J_2 \times J_3}$ reflects the interaction among three distinct factor matrices. Tensor representation can seamlessly exploit all the information in every dimension, which provides direct insights into the nature of the data.

For simplicity, we provide Property 1 and Property 2 for the subsequent optimization in the following.

Property 1: For any third-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ that satisfies the Tucker decomposition, $\mathcal{X}$ can be vectorized as the following form by stacking all the 1-mode vectors of tensor $\mathcal{X}$

$$x = (V_3 \otimes V_2 \otimes V_1)c \tag{2}$$

where x and c denote the lexicographical vectorization of the tensor $\mathcal{X}$ and $\mathcal{C}$, respectively.

Property 2: Given an arbitrary third-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and three matrices V_1, V_2, V_3, the tucker decomposition of tensor $\mathcal{X}$ in formula (1) can be rewritten as 1-mode unfolding form $\mathbf{X}_{(1)} = \mathbf{V}_1 \mathbf{C}_{(1)} (\mathbf{V}_3 \otimes \mathbf{V}_2)^\top$.

Related theoretical derivation of Property 2 can be found in [29]. Given the matrix property as described in [31], any four arbitrary matrices A, B, X, and C, ACB = X, can be rewritten by $(B^T \otimes A)\,\mathrm{vec}(C) = vec(X)$, where vec$(\cdot)$ indicates the vectorization process. By vectorizing the 1-mode unfolding form $\mathbf{X}_{(1)} = \mathbf{V}_1 \mathbf{C}_{(1)} (\mathbf{V}_3 \otimes \mathbf{V}_2)^\top$, we can derive formula (2).

III. Proposed gLGCTD Fusion Model

A. Problem Formulation

The ultimate aim of hyperspectral and multispectral image fusion is to estimate $\mathcal{H}$ of the HR-HSI by utilizing the complementary information from the observation $\mathcal{L}$ of LR-HSI and $\mathcal{M}$ of HR-MSI simultaneously.

The popular existing HSI-MSI fusion methods are based on matrix factorization. Let HR-MSI be the third-order tensor $\mathcal{M} \in \mathbb{R}^{W \times H \times s}$, which is spectrally downsampled with respect to the HR-HSI, where W, H, and s represent the row, column, and band numbers of the HR-MSI, respectively. The LR-HSI is denoted by third-order tensor $\mathcal{L} \in \mathbb{R}^{w \times h \times S}$, which is spatially downsampled from HR-HSI, where w, h, and S represent the row, column, and band numbers of the LR-HSI, respectively. $\mathcal{H} \in \mathbb{R}^{W \times H \times S}$ denotes the desired HR-HSI of the same scene. In traditional matrix-based methods, the fusion models are given by

$$\min_{H_{(3)}} \|L_{(3)} - H_{(3)} S\|_F^2 + \|M_{(3)} - P_3 H_{(3)}\|_F^2 + \varphi(\cdot) \tag{3}$$

where $L_{(3)}$, $M_{(3)}$, and $H_{(3)}$ are the spectral mode (3-mode) unfolding matrices of LR-HSI, HR-MSI, and HR-HSI, respectively. P_3 is the spectral response matrix of the spectral mode, and S denotes the spatial downsampling matrix modeling the point spread function (PSF) and the spatial subsampling process in the hyperspectral sensor. $\varphi(\cdot)$ represents the regularization terms containing some priors. The disadvantages of matrix-based fusion methods are that these methods unfold the

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HSI into matrices, and this unfolding process will break the spatial–spectral structures, making it hard to fully exploit the spectral correlations of the HSI.

In our formulation, the proposed approach hypothesizes that the desired $\mathcal{H}$ of HR-HSI can be reconstructed by estimating a core tensor and multiple-factor matrices along each mode. Formally, we represent the $\mathcal{H}$ of HR-HSI via the Tucker decomposition

$$\begin{aligned} &\mathcal{H} = \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 V_3 \\ &\text{s.t.} V_1^T V_1 = I, \quad V_2^T V_2 = I, \ V_3^T V_3 = I \end{aligned} \tag{4}$$

where $\mathcal{C} \in \mathbb{R}^{W \times H \times S}$ is the core tensor, and $V_1 \in \mathbb{R}^{W \times W}$, $V_2 \in \mathbb{R}^{H \times H}$, and $V_3 \in \mathbb{R}^{S \times S}$ denote the factor matrices of the 1-mode (column mode), 2-mode (row mode), and the 3-mode (spectral mode), respectively.

The acquired $\mathcal{M}$ of HR-MSI is regarded as the spectrally downsampled version of the $\mathcal{H}$ of HR-HSI, that is

$$\mathcal{M} = \mathcal{H} \times_3 P_3. \tag{5}$$

Substituting formula (5) into formula (4), we have an equivalent form represented as

$$\begin{aligned} &\mathcal{M} = \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 P_3 V_3 \\ &\text{s.t.} V_1^T V_1 = I, \quad V_2^T V_2 = I, \ V_3^T V_3 = I. \end{aligned} \tag{6}$$

Assume that there exists matrices P_1 and P_2 such that the spatial downsampling process can be modeled, and the following multilinear compression is formulated:

$$\mathcal{L} = \mathcal{H} \times_1 P_1 \times_2 P_2. \tag{7}$$

The spatial downsampling model in (7) is intuitive and reveals an ingenious connection with tensor mode products. It is assumed that the PSF and the subsampling process of the hyperspectral sensors [26] can be modeled exactly by two matrices along the column mode and the row mode, i.e. $S = (P_2 \otimes P_1)^T$.

Substituting (4) into (5) and (7), the HSI-MSI fusion problem is formulated as the following least-squares optimization procedure:

$$\begin{aligned} \min_{\mathcal{C}, V_1, V_2, V_3} &\|\mathcal{L} - \mathcal{C} \times_1 P_1 V_1 \times_2 P_2 V_2 \times_3 V_3\|_F^2 \\ &+ \|\mathcal{M} - \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 P_3 V_3\|_F^2 \end{aligned} \tag{8}$$

where $\|\cdot\|_F$ denotes the Frobenius norm, and the two terms in formula (8) share the same core tensor $\mathcal{C}$. Note that instead of estimating multiple dictionaries and sparse core tensor coefficients just as the work in [26], the HSI-MSI fusion problem is formulated as the estimation of the shared core tensor and three corresponding factor matrices V_1, V_2, and V_3 along each mode simultaneously. In practice, two separated downsampling matrices P_1 and P_2 in the spatial modes can be approximated in real data set experiments [32].

B. Proposed gLGCTD Fusion Model

In this work, the HSI-MSI fusion problem is to estimate the core tensor $\mathcal{C}$ and corresponding factor matrices along each mode in (8). The fusion model in (8) does not have a unique solution, and it is sensitive to noises and outliers without appropriate regularization terms. Therefore, to accurately estimate the four variables, we need proper regularization terms to constrain the solution domain. The Tucker tensor decomposition is a valid multilinear representation for high-dimensional tensor data, but it fails to take the manifold structures of high-dimensional HR-HSI into account. Moreover, the Tucker decomposition employed in (8) is a global decomposition and takes no account of the local geometry of hyperspectral data. Ignoring complementary local spectral–spatial information cannot effectively preserve the details leading to degraded fusion performances. Therefore, it is rational to introduce a regularization term merging some local submanifold structures of HR-HSI.

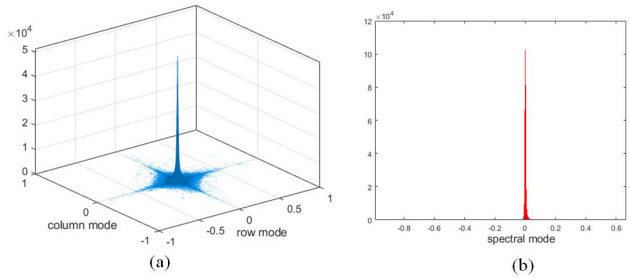


Fig. 2. Statistics of difference values of the University of Pavia data set along (a) spatial modes and (b) spectral mode.

It has been demonstrated that submanifold structures encoded by graph embedding can perfectly preserve local information of high-dimensional data and achieve good performances in many recognition tasks [33], [34]. Different from the existing methods extracting local structural information in the original image domain, in this study, we fully exploit the intrinsic local submanifolds of HR-HSI by seeking for the multilinear embedding space that can preserve the local neighborhood structure by constructing the factor-dependent multilinear graphs. To be exact, the HR-HSI is a third-order tensor with relations spanned by the columns, rows, and spectral modes, and its three latent factors V_1, V_2, and V_3 encode the relations of corresponding spaces. The statistics of difference values along the row, column, and spectral modes of the University of Pavia data set is presented as corresponding histograms in Fig. 2. One can find that most of the difference values are nearly equal to zero from Fig. 2, which justifies the statistical properties of the local similarities existing in all three modes of HR-HSI. The local similarities in three modes can be described as the piecewise smooth structures of corresponding factor matrices in the manifold domain of HR-HSI. The factor matrices V_1, V_2, and V_3 preserve local geometry information of HR-HSI by embedding pixelwise interdependency. As the characteristics of the piecewise smooth structures of factor matrices depend on the topology structure of the graphs on which the factor matrices reside, the topology should be accounted for identifying and exploiting structure in the HSI. By the strength of graph embedding, we extend the graph Laplacian regularizations to interpret the piecewise smooth structures of factor matrices and

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dexterously characterize the latent submanifold structures of HR-HSI in the multilinear space. As such, the local structures of HR-HSI can be disentangled by the topology structure of the graphs characterized by graph Laplacian matrices in the manifold domain to complement the deficiencies of the Tucker decomposition.

That is to say, it is key to construct the p-nearest neighbors (pNN) graphs to characterize the latent submanifold structures of HR-HSI in the multilinear space. From Fig. 2, we infer that adjacent columns/rows/bands are correlated due to the fact that the local similarities of hyperspectral data lie in both intrafactor relations (e.g., among adjacent columns/rows/bands) and interfactor relations (e.g., between spatially adjacent and spectrally adjacent pixels). Therefore, designing the multilinear graphs is an important step to characterize the intrafactor (within V_1, V_2, and V_3) and interfactor (among V_1, V_2, and V_3) relations.

Instead of finding neighbors and building edges between vertices with the same dominated factor (e.g., locality preserving projection [35] and locally linear embedding [36]), which cannot exploit the interfactor relations among different factors, we adopt a factor-dependent strategy to design the multilinear graphs to simultaneously exploit the intrafactor and interfactor relations; more details can be found in [28]. We take the row, column, and spectral factors relations as known *a priori* that posits that samples should share the local similarities within and among row, column, and spectral domains. Given the ith sample $\mathbf{x}_i$ drawn from three corresponding quantized factors (i_1, i_2, i_3), we give the definition of neighboring set $\Omega_k(\mathbf{x}_i)$ in the kth factor by

$$\Omega_k(\mathbf{x}_i) = \{\mathbf{x}_j | j_k \neq i_k, \text{ and } j_{k'} = i_{k'}, \forall k' \neq k\} \tag{9}$$

where (j_1, j_2, j_3) are the multilinear factors of the jth sample $\mathbf{x}_j$. Then, we measure the row, column, and spectral graph edge weight w_{ij}^k as

$$w_{ij}^k = \exp\left(\frac{-|x_i^k - x_j^k|^2}{\sigma^2}\right) \tag{10}$$

where $k \in \{1, 2, 3\}$, and σ determines the smoothing degree. If $\mathbf{x}_j \in pNN(\Omega_k(\mathbf{x}_i))$ or $\mathbf{x}_i \in pNN(\Omega_k(\mathbf{x}_j))$, we measure w_{ij}^k as defined in (10); otherwise, we set $w_{ij}^k = 0$. The $pNN(\Omega_k(\mathbf{x}))$ operator represents the p-nearest neighbors of sample $\mathbf{x}$ within the neighboring set $\Omega_k(\mathbf{x})$. Then, the submanifold structures encoded by the graph embedding framework can be disentangled with the graph Laplacian regularizations as follows:

$$\sum_{k=1}^{3} \text{tr}\left(V_k^T L_k V_k\right) \tag{11}$$

where $V_k (k = 1, 2, 3)$ can be regarded as the kth submanifold, and $L_k (k = 1, 2, 3)$ denominates the so-called Laplacian matrix $L = A - W$. A is a diagonal matrix whose entries are calculated by summing the columns of W and $A_{ii} = \Sigma_j w_{ij}$.

Because the graph Laplacian matrices encode the complementary relations for tensor analysis, the underlying intrafactor and interfactor relations of hyperspectral data should reside on multiple low-dimensional submanifolds. Then, the spatial and spectral submanifold structures are incorporated into a unified framework

$$\begin{aligned}\min_{\mathcal{C}, V_1, V_2, V_3} &\|\mathcal{L} - \mathcal{C} \times_1 P_1V_1 \times_2 P_2V_2 \times_3 V_3\|_F^2 \\ &+ \|\mathcal{M} - \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 P_3V_3\|_F^2 \\ &+ \lambda\|\mathcal{C}\|_F^2 + \delta\sum_{k=1}^{3} \text{tr}\left(V_k^T L_k V_k\right)\end{aligned} \tag{12}$$

where λ and δ are the regularization parameters, and the role of $\|\mathcal{C}\|_F^2$ here is to prevent overfitting and to improve the stability of the coupled tensor decomposition model. Fig. 3 illustrates the proposed gLGCTD fusion model.

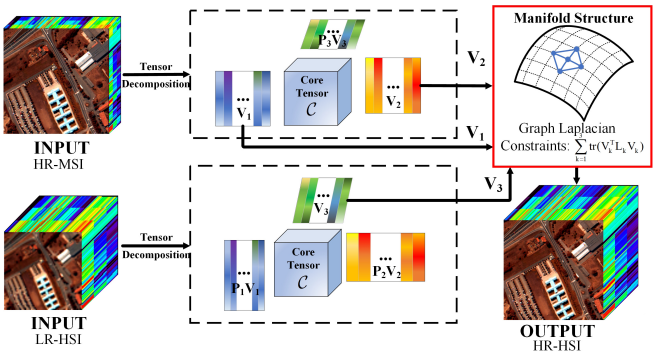


Fig. 3. Illustration of the proposed gLGCTD fusion model.

C. Optimization Algorithm

Optimizing formula (12) is a nonconvex problem by solving $\mathcal{C}$, V_1, V_2, and V_3 jointly, and we can barely obtain the closed-form solutions for $\mathcal{C}$, V_1, V_2, and V_3. However, with respect to each block of variables, the objective function (12) is convex while keeping the other variables fixed. We utilize a new hybrid framework between proximal alternating optimization [37] and the alternating direction method of multipliers [38], [39] to solve the objective function in (12); then, the proposed fusion model can converge to a stationary point of the original NP-hard problem. According to the framework of PAO, the objective function in (12) can be rephrased as four subproblems for $\mathcal{C}$, V_1, V_2, and V_3.

1) Subproblem with respect to $\mathcal{C}$: With the variables V_1, V_2, and V_3 fixed, the optimization with respect to the core tensor $\mathcal{C}$ can be formulated as

$$\begin{aligned}\min_{\mathcal{C}} &\lambda\|\mathcal{C}\|_F^2 + \|\mathcal{L} - \mathcal{C} \times_1 P_1V_1 \times_2 P_2V_2 \times_3 V_3\|_F^2 \\ &+ \|\mathcal{M} - \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 P_3V_3\|_F^2 + \beta\|\mathcal{C} - \mathcal{C}^*\|_F^2\end{aligned} \tag{13}$$

where $\beta > 0$, and $\mathcal{C}^*$ denotes the estimated blocks of variables $\mathcal{C}$ in the previous iteration. Formula (13) can be efficiently solved by the ADMM framework by introducing two splitting variables $\mathcal{C} = \mathcal{C}_1$ and $\mathcal{C} = \mathcal{C}_2$. The unconstrained optimization in (13) can be reformulated as the equivalent constrained form as follows:

$$\begin{aligned}\min_{\mathcal{C}, \mathcal{C}_1, \mathcal{C}_2} &\|\mathcal{L} - \mathcal{C}_1 \times_1 P_1V_1 \times_2 P_2V_2 \times_3 V_3\|_F^2 + \lambda\|\mathcal{C}\|_F^2 \\ \times &\quad \|\mathcal{M} - \mathcal{C}_2 \times_1 V_1 \times_2 V_2 \times_3 P_3V_3\|_F^2 + \beta\|\mathcal{C} - \mathcal{C}^*\|_F^2 \\ \text{s.t. } &\mathcal{C}_1 = \mathcal{C}, \quad \mathcal{C}_2 = \mathcal{C}.\end{aligned} \tag{14}$$

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The optimization in (14), corresponding to a standard ADMM template, can be transformed into the augmented Lagrange function $\mathcal{L}(\mathcal{C}, \mathcal{C}_1, \mathcal{C}_2, \mathcal{A}_1, \mathcal{A}_2)$ as detailed in the following:

$$\begin{aligned}&\mathcal{L}(\mathcal{C}, \mathcal{C}_1, \mathcal{C}_2, \mathcal{A}_1, \mathcal{A}_2)\\ &= \lambda\|\mathcal{C}\|_F^2 + \beta\|\mathcal{C} - \mathcal{C}^*\|_F^2\\ &\quad + \|\mathcal{L} - \mathcal{C}_1 \times_1 P_1V_1 \times_2 P_2V_2 \times_3 V_3\|_F^2 + \mu\|\mathcal{C} - \mathcal{C}_1 - \mathcal{A}_1\|_F^2\\ &\quad + \|\mathcal{M} - \mathcal{C}_2 \times_1 V_1 \times_2 V_2 \times_3 P_3V_3\|_F^2 + \mu\|\mathcal{C} - \mathcal{C}_2 - \mathcal{A}_2\|_F^2\end{aligned} \tag{15}$$

where $\mathcal{A}_1$ and $\mathcal{A}_2$ are the scaled Lagrange multipliers, and μ is the penalty parameter. Hence, the optimization problem in (15) can be split into the following subproblems.

a) Update $\mathcal{C}$: The sub-Lagrange function with respect to $\mathcal{C}$ is formulated as the following by keeping other variables fixed:

$$\begin{aligned}\min_{\mathcal{C}} \beta\|\mathcal{C} - \mathcal{C}^*\|_F^2 + \lambda\|\mathcal{C}\|_F^2 + \mu\|\mathcal{C} - \mathcal{C}_1 - \mathcal{A}_1\|_F^2\\ + \mu\|\mathcal{C} - \mathcal{C}_2 - \mathcal{A}_2\|_F^2.\end{aligned} \tag{16}$$

The closed-form solution to the formula (16) can be easily derived as

$$\mathcal{C} = \frac{\mu}{2\mu + \lambda + \beta}(\mathcal{C}_1 + \mathcal{C}_2 + \mathcal{A}_1 + \mathcal{A}_2) + \frac{\beta}{2\mu + \lambda + \beta}\mathcal{C}^*. \tag{17}$$

b) Update $\mathcal{C}_1$: The sub-Lagrange function with respect to $\mathcal{C}_1$ is formulated as the following by keeping other variables fixed:

$$\min_{\mathcal{C}_1} \|\mathcal{L} - \mathcal{C}_1 \times_1 P_1V_1 \times_2 P_2V_2 \times_3 V_3\|_F^2 + \mu\|\mathcal{C} - \mathcal{C}_1 - \mathcal{A}_1\|_F^2. \tag{18}$$

According to the properties of tensors in formula (2), the abovementioned optimization problem is equivalent to

$$\min_{c_1} \|l - (V_3 \otimes P_2V_2 \otimes P_1V_1)c_1\|_F^2 + \mu\|c - c_1 - \alpha_1\|_F^2 \tag{19}$$

where c, c_1, α_1, and l denote the lexicographical vectorization of the tensor $\mathcal{C}$, $\mathcal{C}_1$, $\mathcal{A}_1$, and $\mathcal{L}$, respectively. The closed-form solution to the formula (19) can be easily derived as

$$\begin{aligned}c_1 = [\mu I + (V_3 \otimes P_2V_2 \otimes P_1V_1)^T(V_3 \otimes P_2V_2 \otimes P_1V_1)]^{-1}\\ \times [(V_3 \otimes P_2V_2 \otimes P_1V_1)^T l + \mu(c - \alpha_1)].\end{aligned} \tag{20}$$

However, the size of matrix $V_3 \otimes P_2V_2 \otimes P_1V_1 \in \mathbb{R}^{whS \times WHS}$ is extremely large with the Kronecker product, and optimization in (20) is complicated to solve. Taking the advantage of matrix diagonalization, we simplify the operand by introducing the unitary matrices U_1', U_2', and U_3' and the diagonal matrices D_1', D_2', and D_3' to rewrite the term $[\mu I + (V_3 \otimes P_2V_2 \otimes P_1V_1)^T(V_3 \otimes P_2V_2 \otimes P_1V_1)]^{-1}$ as $U'(D_3' \otimes D_2' \otimes D_1' + \mu I)^{-1}U'^T$, where $U' = U_3' \otimes U_2' \otimes U_1'$, and D_1', D_2', and D_3' hold the eigenvalues and eigenvectors of $(P_1V_1)^T(P_1V_1)$, $(P_2V_2)^T(P_2V_2)$ and $V_3^TV_3$, respectively. As $(D_3' \otimes D_2' \otimes D_1' + \mu I)^{-1}$ is diagonal, it is simple to calculate. Besides, the term $(V_3 \otimes P_2V_2 \otimes P_1V_1)^T l$ is simplified by vectorizing $\mathcal{L} \times_1 V_1^T P_1^T \times_2 V_2^T P_2^T \times_3 V_3^T$ according to the formula (2). Therefore, we have greatly reduced the computational complexity by such transformation.

c) Update $\mathcal{C}_2$: The sub-Lagrange function with respect to $\mathcal{C}_2$ is formulated as the following by keeping other variables fixed:

$$\min_{\mathcal{C}_2} \|\mathcal{M} - \mathcal{C}_2 \times_1 V_1 \times_2 V_2 \times_3 P_3V_3\|_F^2 + \mu\|\mathcal{C} - \mathcal{C}_2 - \mathcal{A}_2\|_F^2. \tag{21}$$

According to the properties of tensors in formula (2), the abovementioned optimization problem is equivalent to

$$\min_{c_2} \|m - (P_3V_3 \otimes V_2 \otimes V_1)c_2\|_F^2 + \mu\|c - c_2 - \alpha_2\|_F^2 \tag{22}$$

where c, c_2, α_2, and m denote the lexicographical vectorization of the tensor $\mathcal{C}$, $\mathcal{C}_2$, $\mathcal{A}_2$, and $\mathcal{M}$, respectively. The closed-form solution to the formula (22) can be easily derived as

$$\begin{aligned}c_2 = [\mu I + (P_3V_3 \otimes V_2 \otimes V_1)^T(P_3V_3 \otimes V_2 \otimes V_1)]^{-1}\\ \times [(P_3V_3 \otimes V_2 \otimes V_1)^T m + \mu(c - \alpha_2)].\end{aligned} \tag{23}$$

However, the size of matrix $P_3V_3 \otimes V_2 \otimes V_1 \in \mathbb{R}^{WHs \times WHS}$ is extremely large due to the Kronecker product, and it makes the optimization in (23) complicated to solve. Therefore, we take advantage of the properties of matrix diagonalization to simplify the operand. Here, we introduce the unitary matrices U_1, U_2, and U_3 and the diagonal matrices D_1, D_2, and D_3 to rewrite the term $[\mu I + (P_3V_3 \otimes V_2 \otimes V_1)^T(P_3V_3 \otimes V_2 \otimes V_1)]^{-1}$ as $U(D_3 \otimes D_2 \otimes D_1 + (\lambda + \mu)I)^{-1}U^T$, where $U = U_3 \otimes U_2 \otimes U_1$, and D_1, D_2, and D_3 hold the eigenvalues and eigenvectors of $V_1^TV_1$, $V_2^TV_2$, and $(P_3V_3)^T(P_3V_3)$, respectively. Note that $(D_3 \otimes D_2 \otimes D_1 + \mu I)^{-1}$ is diagonal and simple to calculate. Therefore, we can greatly reduce the computational complexity by such transformation. Besides, the term $(P_3V_3 \otimes V_2 \otimes V_1)^T m$ is simplified by vectorizing $\mathcal{M} \times_1 V_1^T \times_2 V_2^T \times_3 V_3^T P_3^T$ according to the formula (2).

d) Update the Lagrangian multipliers $\mathcal{A}_1$ and $\mathcal{A}_2$: After solving each suboptimization problem, we update the Lagrangian multipliers $\mathcal{A}_1$ and $\mathcal{A}_2$ according to the framework of the ADMM by

$$\begin{aligned}\mathcal{A}_1 &\leftarrow \mathcal{A}_1 + \mathcal{C}_1 - \mathcal{C}\\ \mathcal{A}_2 &\leftarrow \mathcal{A}_2 + \mathcal{C}_2 - \mathcal{C}.\end{aligned} \tag{24}$$

2) Subproblem With Respect to V_1: With the variables $\mathcal{C}$, V_2 and V_3 fixed, the optimization with respect to V_1 can be formulated as

$$\begin{aligned}\min_{V_1} \|\mathcal{L} - \mathcal{C} \times_1 P_1V_1 \times_2 P_2V_2 \times_3 V_3\|_F^2 + \beta\|V_1 - V_1^*\|_F^2\\ + \|\mathcal{M} - \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 P_3V_3\|_F^2 + \delta\,\mathrm{tr}\left(V_1^T L_1 V_1\right).\end{aligned} \tag{25}$$

By performing the 1-mode unfolding and substituting the term in (25), we have

$$\begin{aligned}\min_{\mathbf{V}_1} \|L_{(1)} - P_1V_1F_1\|_F^2 + \|M_{(1)} - V_1G_1\|_F^2\\ + \delta\,\mathrm{tr}\left(V_1^T L_1 V_1\right) + \beta\|V_1 - V_1^*\|_F^2\end{aligned} \tag{26}$$

where $L_{(1)}$ represents the 1-mode unfolding of LR-HSI $\mathcal{L}$, $F_1 = (\mathcal{C} \times_2 P_2V_2 \times_3 V_3)_{(1)}$. $M_{(1)}$ represents the 1-mode unfolding of HR-MSI $\mathcal{M}$, $G_1 = (\mathcal{C} \times_2 V_2 \times_3 P_3V_3)_{(1)}$. The optimization problem in (26) is quadratic, and its solution

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is equivalent to compute the following general Sylvester equation:

$$P_1^T P_1 V_1 F_1 F_1^T + V_1\big(G_1 G_1^T + \beta I\big) + 2\delta L_1 V_1 = P_1^T L_{(1)} F_1^T + \beta V_1^* + M_{(1)} G_1^T. \quad (27)$$

Note that the system matrix in (27) satisfies positive definiteness and symmetry; hence, we adopt the conjugate gradient (CG) method [40] to approximate the globally optimal solution for (27).

3) Subproblem With Respect to V_2: With the variables $\mathcal{C}$, V_1 and V_3 fixed, the optimization with respect to V_2 can be formulated as

$$\min_{V_2} \|\mathcal{L} - \mathcal{C} \times_1 P_1 V_1 \times_2 P_2 V_2 \times_3 V_3\|_F^2 + \delta \operatorname{tr}(V_2^T L_2 V_2) + \|\mathcal{M} - \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 P_3 V_3\|_F^2 + \beta \|V_2 - V_2^*\|_F^2. \quad (28)$$

By performing the 2-mode unfolding and substituting the term in (28), we have

$$\min_{V_2} \|L_{(2)} - P_2 V_2 F_2\|_F^2 + \|M_{(2)} - V_2 G_2\|_F^2 + \delta tr\big(V_2^T L_2 V_2\big) + \beta \|V_2 - V_2^*\|_F^2 \quad (29)$$

where $L_{(2)}$ represents the 2-mode unfolding of $\mathcal{L}$ from LR-HSI, $F_2 = (\mathcal{C} \times_1 P_1 V_1 \times_3 V_3)_{(2)}$. $M_{(2)}$ represents the 2-mode unfolding of $\mathcal{M}$ from HR-MSI, and $G_2 = (\mathcal{C} \times_1 V_1 \times_3 P_3 V_3)_{(2)}$. The optimization problem in (29) is quadratic, and its solution is equivalent to compute the following general Sylvester equation:

$$P_2^T P_2 V_2 F_2 F_2^T + V_2\big(G_2 G_2^T + \beta I\big) + 2\delta L_2 V_2 = P_2^T L_{(2)} F_2^T + \beta V_2^* + M_{(2)} G_2^T. \quad (30)$$

Note that the system matrix in (30) satisfies positive definiteness and symmetry; hence, we adopt the CG method [40] to approximate the globally optimal solution for (30).

4) Subproblem With Respect to V_3: With the variables $\mathcal{C}$, V_1 and V_2 fixed, the optimization with respect to V_3 can be formulated as

$$\min_{V_3} \|\mathcal{L} - \mathcal{C} \times_1 P_1 V_1 \times_2 P_2 V_2 \times_3 V_3\|_F^2 + \beta \|V_3 - V_3^*\|_F^2 + \|\mathcal{M} - \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 P_3 V_3\|_F^2 + \delta \operatorname{tr}\big(V_3^T L_3 V_3\big). \quad (31)$$

By performing the 3-mode unfolding and substituting the term in (31), we have

$$\min_{V_3} \|L_{(3)} - V_3 F_3\|_F^2 + \|M_{(3)} - P_3 V_3 G_3\|_F^2 + \delta \operatorname{tr}\big(V_3^T L_3 V_3\big) + \beta \|V_3 - V_3^*\|_F^2 \quad (32)$$

where $L_{(3)}$ represents the 3-mode unfolding of $\mathcal{L}$ from LR-HSI, and $F_3 = (\mathcal{C} \times_1 P_1 V_1 \times_2 P_2 V_2)_{(3)}$. $M_{(3)}$ represents the 3-mode unfolding of $\mathcal{M}$ from HR-MSI, and $G_3 = (\mathcal{C} \times_1 V_1 \times_2 V_2)_{(3)}$. The optimization problem in (32) is quadratic, and its solution is equivalent to compute the following general Sylvester equation:

$$P_3^T P_3 V_3 G_3 G_3^T + V_3(F_3 F_3^T + \beta I) + 2\delta L_3 V_3 = L_{(3)} F_3^T + \beta V_3^* + P_3^T M_{(3)} G_3^T. \quad (33)$$

Algorithm 1 gLGCTD-Based HSI-MSI Fusion

Input: LR-HS image $\mathcal{L}$, HR-MS image $\mathcal{M}$, spatial downsampling matrices P_1 and P_2, spectral response matrix P_3, graph Laplacian matrix $L_i (i = 1, 2, 3)$, $\lambda > 0$, $\delta > 0$, $\beta > 0$ and $\mu > 0$

Output: factor matrices V_1, V_2, V_3 and core tensor $\mathcal{C}$

Initialize: Generate pseudorandom factor matrices V_1, V_2, V_3 and core tensor $\mathcal{C} \in \mathbb{R}^{W \times H \times S}$

While not converged do

1) Solve core tensor $\mathcal{C}$ via ADMM framework

While not converged do

(1). Update core tensor $\mathcal{C}$ via Eq. (17)

(2). Update variable $\mathcal{C}_1$ via Eq. (20)

(3). Update variable $\mathcal{C}_2$ via Eq. (23)

(4). Update Lagrangian multipliers $\mathcal{A}_1$, $\mathcal{A}_1$ via Eq. (24)

End while

2) Solve factor matrices V_1 by Eq. (27) with CG

3) Solve factor matrices V_2 by Eq. (30) with CG

4) Solve factor matrices V_3 by Eq. (33) with CG

End while

Note that the system matrix in (33) satisfies positive definiteness and symmetry; hence, we adopt the CG method [40] to approximate the globally optimal solution for (33).

Algorithm 1 summarizes each step of the graph Laplacian-guided coupled tensor decomposition model.

IV. Experiments and Results

In this section, the performances of the proposed gLGCTD fusion method are demonstrated on several synthetic and real data sets. The data sets and the quantitative metrics are introduced first. Then, experimental results on both synthetic and real data sets are present, and the effectiveness of the proposed graph Laplacian constraints is discussed. At last, the parameters' sensitivity, computational complexity, and convergence are analyzed.

A. Experimental Results on Synthetic Data

Two open source hyperspectral data sets are used:

1) University of Pavia Data Set [41]: This data set was collected by the reflective optics system imaging spectrometer (ROSIS) optical sensor over the University of Pavia, Italy. It originally consists of 115 bands, which is reduced to 93 bands after removing the water vapor absorption and heavy noise bands. Despite the full size of the University of Pavia data is $610 \times 340 \times 115$, we extract the top left 256×256 subimage for the experiment and the left bottom 256×256 subimage used as the reference images to facilitate the downsampling process. The original HR-HSI is degraded via averaging the $s \times s$ disjoint spatial blocks, where s denotes the spatial downsampling factor, to generate the simulated LR-HSI. The HR-MSI of four bands is generated by an IKONOS-like spectral reflectance response filter [42].

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2) CAVE Data Set [43]: This data set was captured by a generalized assorted pixel camera, consisting of 32 indoor HSIs with a size of $512 \times 512 \times 31$. The spectral wavelength is in the range of 400–700 nm, and the interval is 10 nm. The spectral downsampling matrix, which stems from the response of the Nikon D700 camera, is adopted to simulate the HR-MSI via downsampling the HR-HSI along the spectral mode. Same as the Pavia data set, the LR-HSI is generated by averaging the $s \times s$ disjoint spatial blocks of the HR-HSI due to the separability of the spatial degeneration operators.

Five quantitative metrics are used to evaluate the quality of the reconstructed HSI. As defined in [26], root-mean-square error (RMSE) and the peak signal-to-noise ratio (PSNR) are used to evaluate the quality of the spatial structure of reconstructed HSI; spectral angle mapper (SAM) [18] and degree of distortion (DD) assess spectral distortions; and the relative dimensionless global error in synthesis (ERGAS) [44] measures the overall quality of the fused HSI.

The proposed gLGCTD method is compared with five state-of-the-art hyperspectral and multispectral image fusion approaches on the abovementioned hyperspectral data sets, including Generalization of Simultaneous Orthogonal Matching Pursuit (G-SOMP+) [16], group spectral embedding (GSE) [45], Bayesian sparse representation (BSR) [46], coupled spectral unmixing (CSU) [14], and coupled sparse tensor factorization (CSTF) [26]. We have tested the downsampling factors $s = 4, s = 8,$ and $s = 16$. All the experiments are implemented using MATLAB R2018b on the computer with Core i7-7700 CPU with 3.60 GHz and 64-GB RAM.

The performances for the University of Pavia data set with a downsampling factor 8 are presented in Figs. 4 and 5. Fig. 4 shows the reconstructed HSI at the 55th band, and Fig. 5 shows the reconstructed HSI at the 30th band. From Figs. 4 and 5, the fine and sharp spatial structures and textures of the HSI can be well reconstructed by all the fusion methods except the GSE method. It is due to the fact that the spatial downsampling matrix is not required as prior information in the GSE method, which is generally unknown before and needs to be accurately estimated in practice. Obviously, the proposed gLGCTD method outperforms all other compared fusion methods in recovering the spatial and spectral details of the original HSI.

The averages of the RMSE, PSNR, SAM, DD, and ERGAS results of the compared methods with downsampling factors 4, 8, and 16 for the University of Pavia data sets are reported in Tables I and II. The best values are annotated in bold. From Tables I and II, we can observe that the performance of GSE is the worst, and it is ineffective in recovering the fine spatial details. Although the CSTF method is relatively better than the other fusion methods, the PSNR is 1.1 dB lower than the proposed gLGCTD method. GSE and G-SOMP+ both attain low PSNR values and high SAM values, which means the spatial and spectral structures are not well recovered. On average, for all quantitative metrics, RMSE, PSNR, SAM, DD, and ERGAS performances of the proposed gLGCTD method are comparable or better than those of other fusion methods. The fine spatial structures and spectral characteristics are preserved by using the graph Laplacian-guided coupled tensor decomposition.

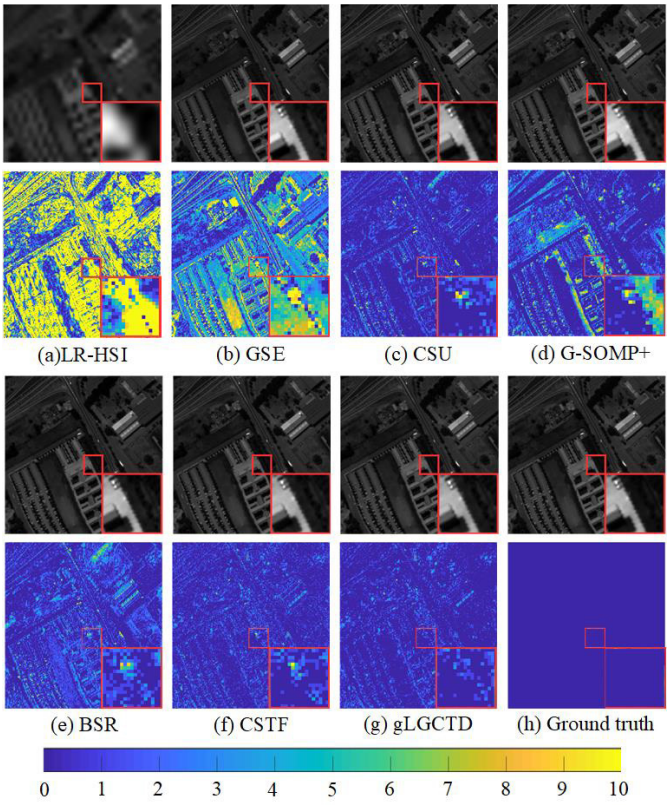


Fig. 4. Reconstructed HSIs and corresponding error images of the 55th band with a downsampling factor 8 for University of Pavia data 1(top left 256×256 pixels). (a) LR-HSI. (b) GSE [45]. (c) CSU [14]. (d) G-SOMP+ [16]. (e) BSR [46]. (f) CSTF [26]. (g) Proposed gLGCTD. (h) Ground truth.

Fig. 6 shows the average PSNR curves as a function of the spectral bands over the University of Pavia data set for all the compared fusion methods with different downsampling factors. Whether the downsampling factors are 2, 4, 8, or 16, we can observe that the proposed gLGCTD method outperforms other methods consistently at each spectral band. This further illustrates the ability of gLGCTD in spectral preservation.

Same as the experimental settings in the University of Pavia data set, we test the CAVE hyperspectral images with downsampling factors $s = 2, 4, 8,$ and 16. Due to the numerous hyperspectral images in the CAVE data set, we utilize one representative image in the CAVE data set with downsampling factors for $s = 2$ and $s = 4$, as shown in Fig. 7. We also conduct experiments using all the CAVE hyperspectral images (containing 32 images) to validate the effectiveness of the proposed gLGCTD method with downsampling rates of 8 and 16. For visual comparison, a part of the fused hyperspectral images at 440 and 530 nm by the compared fusion methods is presented in Figs. 8 and 9, respectively. Both GSE and G-SOMP+ have produced oversmooth reconstruction, and the spatial details and structures are not well characterized. Although all the remaining compared fusion methods have generated satisfactory results, it is shown that the proposed gLGCTD method can well recover the spatial details and structures at the smallest reconstruction errors.

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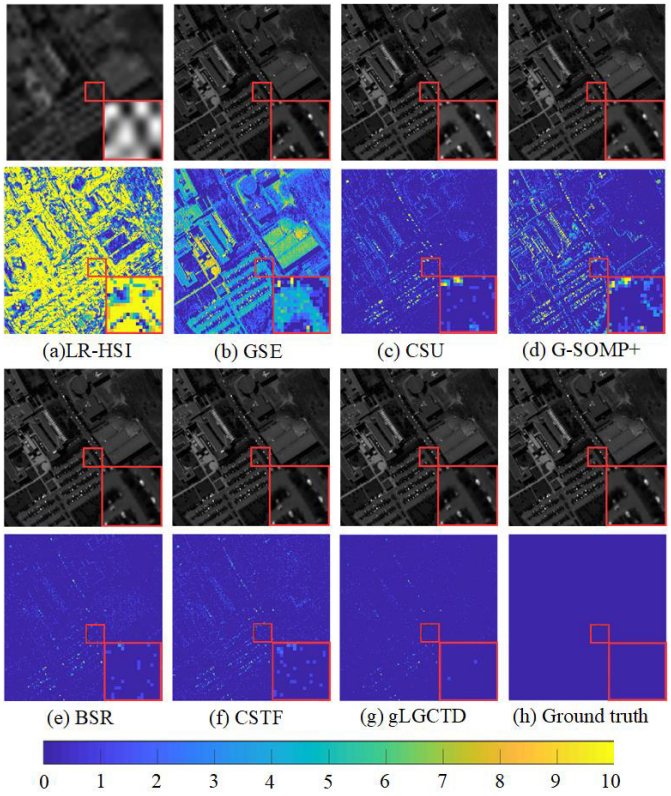


Fig. 5. Reconstructed HSIs and corresponding error images of the 30th band with a downsampling factor 8 for University of Pavia data 2 (bottom left 256×256 pixels). (a) LR-HSI. (b) GSE [45]. (c) CSU [14]. (d) G-SOMP+ [16]. (e) BSR [46]. (f) CSTF [26]. (g) Proposed gLGCTD. (h) Ground truth.

TABLE I

AVERAGES OF RMSE, PSNR, SAM, DD, AND ERGAS RESULTS OF THE COMPARED METHODS FOR DIFFERENT DOWNSAMPLING FACTORS ON THE UNIVERSITY OF PAVIA DATA 1

Method	GSE	CSU	G-SOMP+	BSR	CSTF	**Proposed gLGCTD**
			s=2			
RMSE	4.68	1.66	2.54	2.05	1.47	**1.31**
PSNR	34.94	44.18	41.04	43.21	45.43	**46.65**
SAM	2.63	1.48	2.50	2.02	1.56	**1.37**
DD	3.45	0.86	1.58	1.26	0.93	**0.77**
ERGAS	5.27	1.98	2.85	2.23	1.65	**1.46**
			s=4			
RMSE	4.89	1.95	2.65	2.01	1.71	**1.57**
PSNR	34.54	42.88	40.56	43.33	44.38	**45.34**
SAM	2.71	1.72	2.55	2.02	1.77	**1.60**
DD	3.64	1.04	1.63	1.24	1.08	**0.93**
ERGAS	2.75	1.11	1.48	1.10	0.95	**0.85**
			s=8			
RMSE	4.95	2.18	2.80	2.13	1.86	**1.73**
PSNR	34.46	41.90	40.04	42.53	43.62	**44.53**
SAM	2.95	1.92	2.53	2.15	1.93	**1.77**
DD	3.56	1.20	1.65	1.34	1.18	**1.06**
ERGAS	1.40	0.61	0.79	0.60	0.52	**0.47**
			s=16			
RMSE	6.17	2.36	4.13	2.23	2.34	**1.94**
PSNR	32.65	41.22	36.58	42.57	41.90	**43.77**
SAM	2.45	2.06	2.84	2.12	2.35	**2.01**
DD	4.86	1.31	2.12	1.33	1.46	**1.20**
ERGAS	1.66	0.67	1.19	0.61	0.67	**0.26**

The averages of the RMSE, PSNR, SAM, DD, and ERGAS results of the 32 reconstructed HSIs for the compared methods with different downsampling factors on the CAVE database

TABLE II

AVERAGES OF RMSE, PSNR, SAM, DD, AND ERGAS RESULTS OF THE COMPARED METHODS FOR DIFFERENT DOWNSAMPLING FACTORS ON THE UNIVERSITY OF PAVIA DATA 2

Method	GSE	CSU	G-SOMP+	BSR	CSTF	**Proposed gLGCTD**
			s=2			
RMSE	5.13	1.96	3.33	2.15	1.69	**1.45**
PSNR	34.15	42.78	38.39	42.67	44.25	**45.60**
SAM	3.02	1.64	2.79	3.11	1.83	**1.54**
DD	3.84	0.85	1.72	1.88	0.99	**0.78**
ERGAS	5.95	2.43	4.02	3.21	2.01	**1.77**
			s=4			
RMSE	5.17	2.56	3.56	2.33	1.91	**1.76**
PSNR	34.08	40.39	37.60	41.93	43.36	**44.22**
SAM	3.08	1.95	3.11	2.26	2.01	**1.80**
DD	3.85	1.09	1.97	1.22	1.09	**0.94**
ERGAS	2.99	1.55	2.09	1.22	1.11	**1.02**
			s=8			
RMSE	5.21	2.75	4.03	2.37	2.15	**1.92**
PSNR	34.07	39.69	36.43	41.60	42.37	**43.52**
SAM	2.83	2.06	3.07	2.35	2.21	**1.95**
DD	3.98	1.18	1.83	1.31	1.23	**1.04**
ERGAS	1.47	0.83	1.23	0.70	0.64	**0.55**
			s=16			
RMSE	7.67	3.20	4.88	3.41	2.31	**2.04**
PSNR	30.96	38.35	34.72	40.62	41.79	**43.08**
SAM	2.49	2.40	3.74	2.42	2.36	**2.08**
DD	6.27	1.41	2.29	1.34	1.33	**1.12**
ERGAS	1.02	0.50	0.73	0.33	0.35	**0.29**

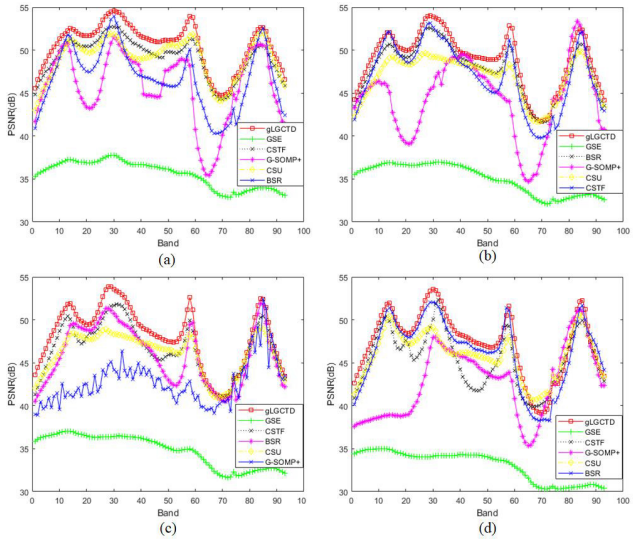


Fig. 6. Average PSNR curves as a function of the spectral bands for the compared approaches on (a)–(d) University of Pavia data set with downsampling factors 2, 4, 8, and 16, respectively.

are listed in Table III. The best values are annotated in bold. Although the spatial details and structures are well preserved visually by the CSU method, the numerical quality results with respect to spectral recovery are not promising. The spatial and spectral details obtained by the CSTF method are not well preserved compared with the proposed gLGCTD because CSTF regards the whole HSI as a 3-D low-rank tensor and the local manifold information of the hyperspectral scene is not fully exploited. We can easily notice that the proposed

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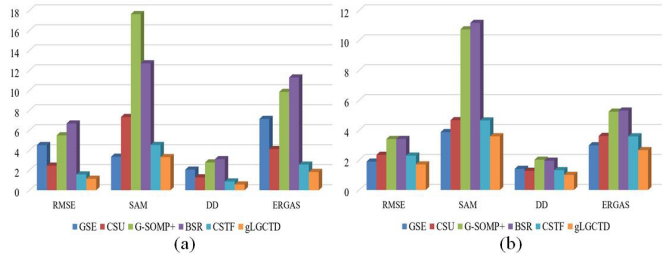


Fig. 7. Some quantitative evaluation results of the compared methods on the CAVE data set with different downsampling factors. (a) *feathers_ms* with downsampling factor $s = 2$. (b) *oil_painting_ms* with downsampling factor $s = 4$.

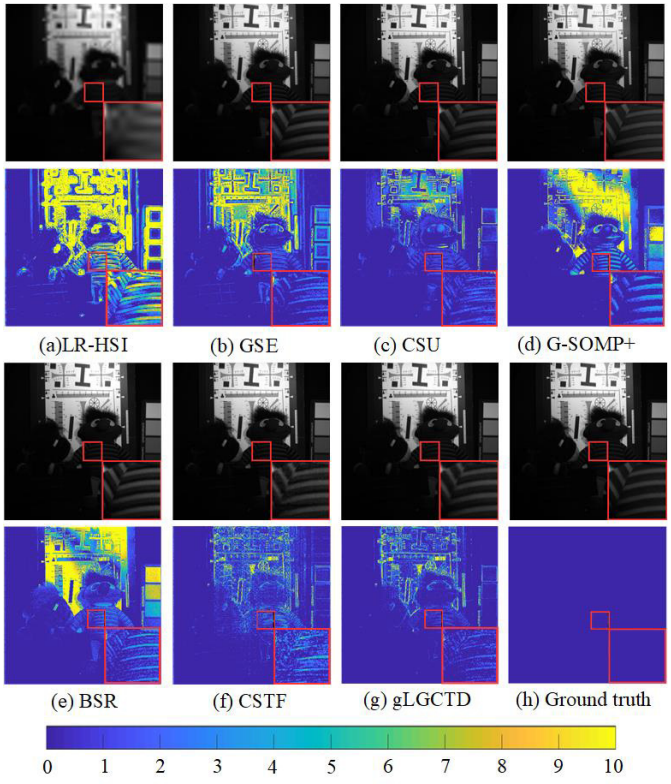


Fig. 8. Reconstructed HSIs and corresponding error images of *chart_and_stuffed_toy_ms* in the CAVE data set at 440 nm with a downsampling factor 8. (a) LR-HSI. (b) GSE [45]. (c) CSU [14]. (d) G-SOMP+ [16]. (e) BSR [46]. (f) CSTF [26]. (g) Proposed gLGCTD. (h) Ground truth.

gLGCTD method still outperforms other compared methods with well preserving spatial and spectral details.

Fig. 10 shows the average PSNR curves as a function of the spectral bands over the CAVE data sets for all the compared fusion methods with different downsampling factors. At both downsampling factors 8 or 16, the proposed gLGCTD method outperforms other methods consistently at almost all spectral bands.

B. Effectiveness of the Proposed Graph Laplacian Constraints

To further verify that the proposed gLGCTD can preserve the useful spectral and spatial information of the HSI, we give a visual comparison between the reconstructed HSI with and without spatial–spectral factor graph constraints in Fig. 11. Fig. 11(b) shows the result of reconstructed HSI with only spectral factor graph constraint. Fig. 11(c) shows the result of reconstructed HSI with only spatial factor graph constraint. Fig. 11(d) shows the result of the proposed gLGCTD with both spectral and spatial factor graph constraints. It can be inferred from the PSNR value and visual performance that both spectral and spatial factor graph constraints have an obvious advantage over only spectral factor graph constraint in terms of texture preserving. Moreover, not only both spectral and spatial factor

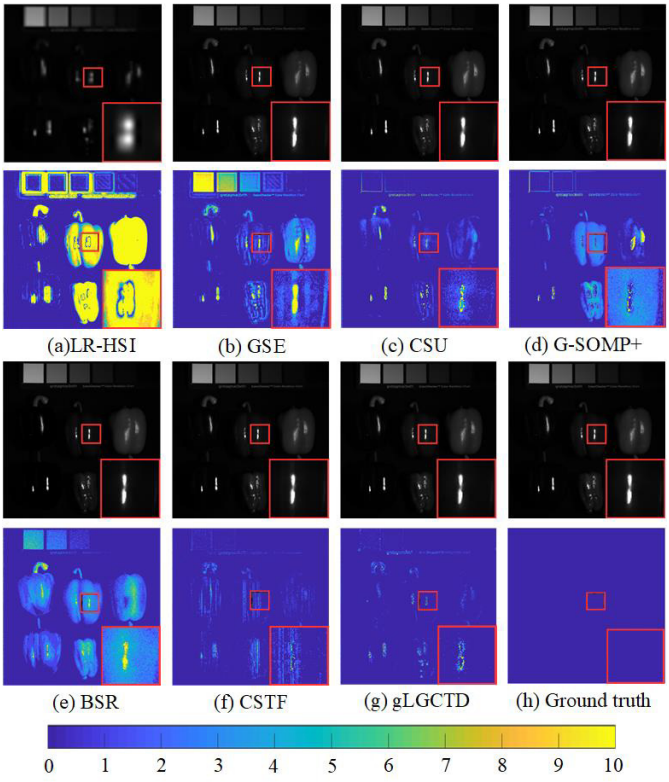


Fig. 9. Reconstructed HSIs and corresponding error images of *peppers_ms* in CAVE data set at 530 nm with a downsampling factor 8. (a) LR-HSI. (b) GSE [45]. (c) CSU [14]. (d) G-SOMP+ [16]. (e) BSR [46]. (f) CSTF [26]. (g) Proposed gLGCTD. (h) Ground truth.

TABLE III

AVERAGES OF RMSE, PSNR, SAM, DD, AND ERGAS RESULTS OF THE COMPARED METHODS FOR DIFFERENT DOWNSAMPLING FACTORS ON THE CAVE DATA SET [43]

Method	GSE	CSU	G-SOMP+	BSR	CSTF	**Proposed gLGCTD**
s=8						
RMSE	3.44	2.49	4.02	4.86	2.24	**1.66**
PSNR	37.76	42.20	39.35	37.98	43.52	**45.34**
SAM	6.45	6.63	13.65	10.63	6.95	**5.15**
DD	1.89	1.13	1.82	1.30	1.21	**0.82**
ERGAS	1.10	1.40	1.62	1.62	1.20	**0.95**
s=16						
RMSE	3.76	2.74	4.58	5.24	2.59	**2.12**
PSNR	37.06	41.24	38.70	37.56	42.57	**43.67**
SAM	6.84	6.68	19.36	11.36	7.60	**6.30**
DD	1.97	1.23	2.30	2.41	1.37	**1.05**
ERGAS	1.25	0.78	1.14	2.62	0.67	**0.60**

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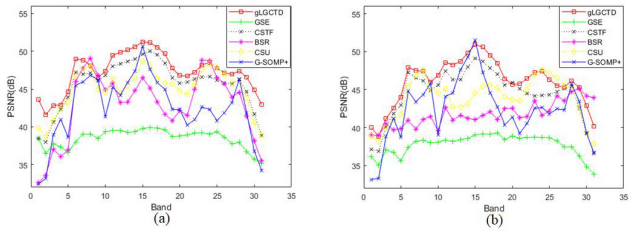


Fig. 10. Average PSNR curves as a function of the spectral bands for the compared approaches on (a) and (b) CAVE data set with downsampling factors 8 and 16, respectively.

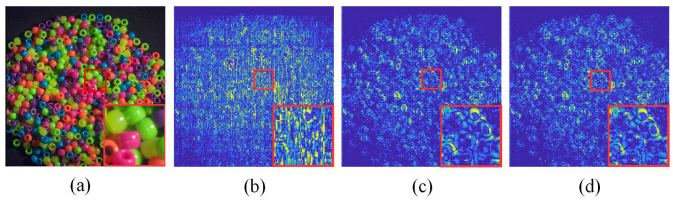


Fig. 11. Effectiveness of the proposed graph Laplacian constraints on *beads_ms* in CAVE database. (a) Original image. (b) Reconstructed result with only spectral factor graph constraint (PSNR = 34.94 dB, and SAM = 10.79). (c) Reconstructed result with only spatial factor graph constraint (PSNR = 36.97 dB, and SAM = 6.80). (d) Reconstructed result with both spectral and spatial factor graph constraints (PSNR = 37.13 dB, and SAM = 6.68).

graph constraints outperform the visual performance with only spatial factor graph constraint but also effectively reduce the spectral distortions on account of smaller SAM value.

C. Parameters' Sensitivity Discussions

We have investigated the sensitivity of the regularization parameters δ and λ and how they influence the fusion quality of the proposed gLGCTD fusion method. In our analysis, δ is chosen from $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}\}$, and λ is chosen from $\{10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}\}$. Fig. 12(a) shows the PSNR values of the fused University of Pavia and CAVE data sets as a function of $\log(\delta)$. It is observable that the PSNR values for the University of Pavia data set and CAVE database increase when $\log(\delta) = [-5, 0]$ and $\log(\delta) = [-5, -2]$ and then decrease sharply for $\log(\delta) > 0$ and $\log(\delta) > -2$. Thus, δ is eventually set as 1 and 0.01 to achieve the highest PSNR values for the University of Pavia data set and CAVE database, respectively. Fig. 12(b) shows the PSNR values of the fused University of Pavia and CAVE data sets as a function of $\log(\lambda)$. For the University of Pavia data set, the PSNR values keep increasing when $\log(\lambda) = [-7, -3]$ and decrease when $\log(\lambda) > -3$. Thus, λ is eventually set as 0.001 for the University of Pavia data set. For the CAVE database, the PSNR values keep increasing when $\log(\lambda) = [-7, -4]$ and then decline with fluctuation when $\log(\lambda) > -4$. Thus, in our experiments, we eventually set λ as 0.0001 for the CAVE database.

D. Experimental Results on Real Data

Thus, to further evaluate the effectiveness of the proposed method, the gLGCTD is subsequently applied to real HSI and MSI fusion. The real LR-HSI data set is collected by the Hyperion sensor onboard of Earth Observing-1 satellite, which has 220 spectral bands in the wavelength regions of 0.4–2.5 μm and the spatial resolution of 30 m; 89 bands are preserved after removing the bands that suffer from the noise. An area of size $210 \times 210 \times 89$ is used in the subsequent experiment. The real HR-MSI data set is captured by the Sentinel-2 satellite, which has 13 spectral bands. Four bands with the spatial resolution of 10 m are utilized for the data fusion. To be exact, four bands correspond to the central wavelength of 490, 560, 665, and 842 nm, respectively. The spatial downsampling factor is 3, and the used Sentinel-2 MSI is of size $630 \times 630 \times 4$. The spatial downsampling matrix and spectral response matrix are estimated using the method in [19]. The first and second rows of Fig. 13(a)–(f) show the reconstructed HR-HSI at the first and 20th bands by different methods. As shown in the figure, all the methods can improve the spatial resolution of the Hyperion LR-HSI to a different extent. The methods of G-SOMP+ and GSE have oversmooth results compared with the others. There are some artificial effects existing in the result of CSTF because the spatial downsampling matrix is approximated in two spatial modes that will cause some extra errors. The spatial details can be well reconstructed by the other fusion methods. The abovementioned experiment demonstrates that the proposed method gLGCTD can perform well in the fusion of real data.

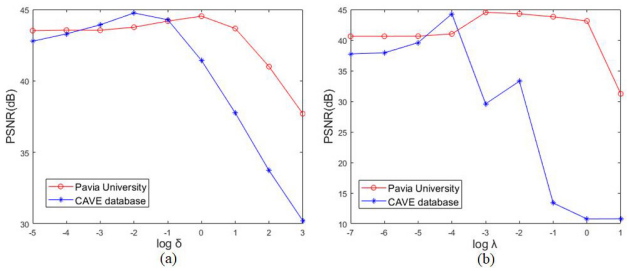


Fig. 12. PSNR curves for the proposed gLGCTD method on the University of Pavia data set and the CAVE database as functions of regularization parameters δ and λ, respectively.

E. Computational Complexity Analysis of gLGCTD

Algorithm 1 is built on the hybrid framework between PAO and ADMM scheme by iteratively solving the factor matrices V_1, V_2, and V_3 and core tensor $\mathcal{C}$ until finding the local optima. The main computational complexity of gLGCTD is the construction of spatial and spectral factor graphs and the optimization procedure. Thereinto, we need $\mathcal{O}(Wp+Hp+Sp)$ to construct the p-nearest neighbor graphS in the initialization.

Regarding the iterative optimization of the conjugate gradient method with respect to (27), (30), and (33), the most time-consuming are the multiplication of the linear system matrix and a vector in the process of optimizing factor matrices V_1, V_2, and V_3, whose computational complexities are $\mathcal{O}(W^4)$, $\mathcal{O}(H^4)$, and $\mathcal{O}(S^4)$, respectively. By matrix representation, the computational complexities can be reduced to $\mathcal{O}(W^3)$, $\mathcal{O}(H^3)$, and $\mathcal{O}(S^3)$ in each iteration. Moreover, optimizing the core tensor $\mathcal{C}$ involves the computation of matrix inversion

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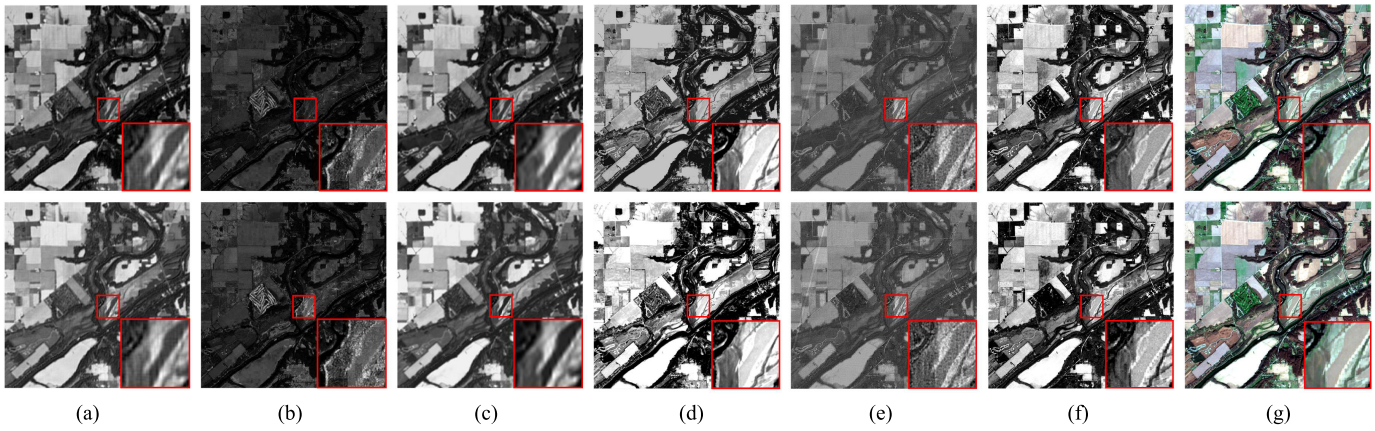


Fig. 13. First and second rows show the reconstructed HR-HSI at first and 20th bands by different methods. (a) CSTF. (b) G-SOMP+. (c) GSE. (d) CSU. (e) BSR. (f) gLGCTD. (g) Sentinel-2 HR-MSI.

in (20) and (23). Without leveraging with diagonalization, the computational complexity can reach $\mathcal{O}(W^3H^3S^3)$. It is reduced to $\mathcal{O}(W^2HS + WH^2S + WHS^2)$ via circumventing the operation of matrix inversion. Therefore, the holistic computational complexities are

$$\mathcal{O}(Wp + Hp + Sp) + \mathcal{O}(T_{\text{CG}}(W^3 + H^3 + S^3)) + \mathcal{O}(T_{\text{ADMM}}(W^2HS + WH^2S + WHS^2)) \quad (34)$$

where T_{CG} and T_{ADMM} denote the iterations of the conjugate gradient and the ADMM, respectively.

F. Convergence Analysis

Algorithm 1 summarizes the flowchart of the hybrid framework between PAO and ADMM by estimating factor matrices V_1, V_2, and V_3 and core tensor $\mathcal{C}$, which aims at solving problem (12). The following proposition demonstrates the convergence of core tensor $\mathcal{C}$ by the ADMM framework in Algorithm 1.

proposition 1: For any β, $\lambda \geq 0$, and $\mu > 0$, the core tensor $\mathcal{C}$ optimizing by ADMM in Algorithm 1 converges to the solution of the objective function in (13). For the case $\beta > 0$, the objective function in (13) has a unique solution.

Proof: Each term of objective function in (13) is proper, closed, and convex function with domain $\mathbb{R}^{WHS}$

$$f_1(\mathcal{C}) = \beta\|\mathcal{C} - \mathcal{C}^*\|_F^2 + \lambda\|\mathcal{C}\|_F^2 \quad (35)$$

$$f_2(\mathcal{C}_1) = \|\mathcal{L} - \mathcal{C}_1 \times_1 P_1V_1 \times_2 P_2V_2 \times_3 V_3\|_F^2 \quad (36)$$

$$f_3(\mathcal{C}_2) = \|\mathcal{M} - \mathcal{C}_2 \times_1 V_1 \times_2 V_2 \times_3 P_3V_3\|_F^2. \quad (37)$$

The functions f_1 and $f_2 + f_3$ are also proper, closed, and convex. The linear mappings $\mathbb{R}^{WHS} \to \mathbb{R}^{2\ WHS}$ of null space are zero vectors. When $\beta > 0$ or $\lambda > 0$, the objective function is the strictly convex function and has the unique solution; when $\beta = \lambda = 0$, since it is the sum of semipositive quadratic parts, the objective function in (13) has solutions; and when $\mu > 0$, the sequences of core tensor $\mathcal{C}$ converge by invoking Theorem 8 in [38]. ■

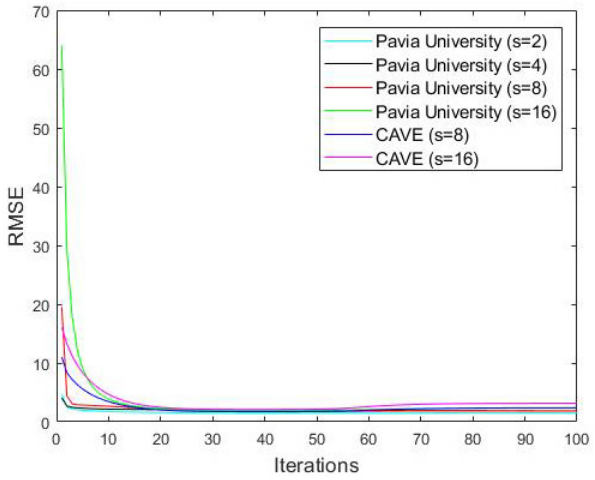


Fig. 14. Empirical convergence analysis of the proposed gLGCTD method.

For convenience, we rewrite the problem (13) for proving the convergence of Algorithm 1 under PAO scheme as follows:

$$G(\mathcal{C}, V_1, V_2, V_3) = \Phi(\mathcal{C}, V_1, V_2, V_3) + \zeta(V_1, V_2, V_3) + \psi(\mathcal{C}) \quad (38)$$

where

$$\begin{cases} \Phi(\mathcal{C}, V_1, V_2, V_3) = \|\mathcal{L} - \mathcal{C} \times_1 P_1V_1 \times_2 P_2V_2 \times_3 V_3\|_F^2 \\ +\|\mathcal{M} - \mathcal{C} \times_1 V_1 \times_2 V_2 \times_3 P_3V_3\|_F^2 \\ \zeta(V_1, V_2, V_3) = \delta\sum_{k=1}^{3} \text{tr}\left(V_k^T L_k V_k\right) \\ \psi(\mathcal{C}) = \lambda\|\mathcal{C}\|_F^2. \end{cases} \quad (39)$$

The function Φ is C^1 with the Lipschitz continuous gradient, namely, consequences of the boundness of $(\mathcal{C}^t, V_1^t, V_2^t, V_3^t)_{t\in\mathbb{N}}$. The functions ζ and ψ are proper, closed, convex, and bounded below. Under the abovementioned conditions, we may invoke the Proposition 1 in [47] that if the sequences $(\mathcal{C}^t, V_1^t, V_2^t, V_3^t)_{t\in\mathbb{N}}$ generated by the PAO+ADMM in Algorithm 1 are bounded. Then, it converges to some stationary point of G.

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Fig. 14 gives an empirical analysis of the convergence of the proposed gLGCTD fusion method. The evolutional curves of RMSE values versus the iterations on the University of Pavia data set and CAVE database with different dowmsampling factors are illustrated. It is easy to observe that all the RMSE curves decrease monotonically and rapidly to the stable values in a few iterations. The algorithm is always converging in just a few iterations (empirically 20 steps) in the implementation.

V. Conclusion

In this article, a novel graph Laplacian-guided coupled tensor decomposition model is proposed for tackling the fusion of HSI-MSI. In order to capture the global interdependencies across the different modes of hyperspectral data, the coupled Tucker decomposition is employed to exploit jointly and seamlessly the intrinsic spatial–spectral information of LR-HSI and HR-MSI. Considering the high-dimensional characteristics of hyperspectral data, we further incorporate the complementary submanifold structures embedded in HR-HSI into the coupled Tucker decomposition framework to locally preserve the underlying spatial–spectral information. The local submanifold structures of HSI encoded by the graph Laplacian regularization and global spatial–spectral information captured by the coupled Tucker decomposition are incorporated into a unified framework. Furthermore, an efficient graph Laplacian-guided coupled tensor decomposition algorithm is proposed using a hybrid framework between PAO and ADMM to produce a more accurate reconstruction of HR-HSI. Experimental results and in-depth discussions on several synthetic and real hyperspectral data sets demonstrate that the proposed gLGCTD method can attain smaller reconstruction errors and finer visual effects than state-of-the-art HSI-MSI fusion methods.

References

[1] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[2] J. H. Wen, Y. Q. Zhao, X. F. Zhang, W. D. Yan, and W. Lin, "Local discriminant non-negative matrix factorization feature extraction for hyperspectral image classification," *Int. J. Remote Sens.*, vol. 35, no. 13, pp. 5073–5093, Jul. 2014.

[3] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[4] J. Liang, J. Zhou, X. Bai, and Y. Qian, "Salient object detection in hyperspectral imagery," in *Proc. IEEE Int. Conf. Image Process.*, Sep. 2013, pp. 2393–2397.

[5] K. Wu, G. Xu, Y. Zhang, and B. Du, "Hyperspectral image target detection via integrated background suppression with adaptive weight selection," *Neurocomputing*, vol. 315, pp. 59–67, Nov. 2018.

[6] J. Yang, Y.-Q. Zhao, J. C.-W. Chan, and S. G. Kong, "Coupled sparse denoising and unmixing with low-rank constraint for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1818–1833, Mar. 2016.

[7] J. Xue, Y. Zhao, W. Liao, and S. G. Kong, "Joint spatial and spectral low-rank regularization for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1940–1958, Apr. 2018.

[8] J. Xue, Y. Zhao, W. Liao, and J. C.-W. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019.

[9] Y. Zhao, J. Yang, and J. C.-W. Chan, "Hyperspectral imagery super-resolution by spatial–spectral joint nonlocal similarity," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2671–2679, Jun. 2014.

[10] C. Yi, Y.-Q. Zhao, and J. C.-W. Chan, "Hyperspectral image super-resolution based on spatial and spectral correlation fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4165–4177, Jul. 2018.

[11] C. Yi, Y.-Q. Zhao, J. Yang, J. C.-W. Chan, and S. G. Kong, "Joint hyperspectral superresolution and unmixing with interactive feedback," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3823–3834, Jul. 2017.

[12] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[13] R. Kawakami, Y. Matsushita, J. Wright, M. Ben-Ezra, Y.-W. Tai, and K. Ikeuchi, "High-resolution hyperspectral imaging via matrix factorization," in *Proc. CVPR*, Jun. 2011, pp. 2329–2336.

[14] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 3586–3594.

[15] Q. Wei, J. Bioucas-Dias, N. Dobigeon, J.-Y. Tourneret, M. Chen, and S. Godsill, "Multiband image fusion based on spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7236–7249, Dec. 2016.

[16] N. Akhtar, F. Shafait, and A. Mian, "Sparse spatio-spectral representation for hyperspectral image super-resolution," in *Proc. Eur. Conf. Comput. Vis.* Springer, 2014, pp. 63–78.

[17] W. Dong *et al.*, "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.

[18] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

[19] M. Simoes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[20] L. Zhang, W. Wei, C. Bai, Y. Gao, and Y. Zhang, "Exploiting clustering manifold structure for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5969–5982, Dec. 2018.

[21] C. Yi, Y.-Q. Zhao, and J. C.-W. Chan, "Spectral super-resolution for multispectral image based on spectral improvement strategy and spatial preservation strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9010–9024, Nov. 2019.

[22] C. I. Kanatsoulis, X. Fu, N. D. Sidiropoulos, and W.-K. Ma, "Hyperspectral super-resolution: A coupled tensor factorization approach," *IEEE Trans. Signal Process.*, vol. 66, no. 24, pp. 6503–6517, Dec. 2018.

[23] K. Zhang, M. Wang, S. Yang, and L. Jiao, "Spatial–spectral-graph-regularized low-rank tensor decomposition for multispectral and hyperspectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1030–1040, Apr. 2018.

[24] R. Dian, L. Fang, and S. Li, "Hyperspectral image super-resolution via non-local sparse tensor factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5344–5353.

[25] H. Li, W. Li, G. Han, and F. Liu, "Coupled tensor decomposition for hyperspectral pansharpening," *IEEE Access*, vol. 6, pp. 34206–34213, 2018.

[26] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[27] F. Nie, Z. Wang, R. Wang, and X. Li, "Submanifold-preserving discriminant analysis with an auto-optimized graph," *IEEE Trans. Cybern.*, pp. 1–14, 2019.

[28] Y.-L. Chen and C.-T. Hsu, "Multilinear graph embedding: Representation and regularization for images," *IEEE Trans. Image Process.*, vol. 23, no. 2, pp. 741–754, Feb. 2014.

[29] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev.*, vol. 51, no. 3, pp. 455–500, Aug. 2009.

[30] A. Cichocki *et al.*, "Tensor decompositions for signal processing applications: From two-way to multiway component analysis," *IEEE Signal Process. Mag.*, vol. 32, no. 2, pp. 145–163, Mar. 2015.

[31] L. De Lathauwer, B. De Moor, and J. Vandewalle, "On the best rank-1 and rank-($R1$, $R2$, ..., RN) approximation of higher-order tensors," *SIAM J. Matrix Anal. Appl.*, vol. 21, no. 4, pp. 1324–1342, Jan. 2000.

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[32] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Nonlocal patch tensor sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 3034–3047, Jun. 2019.

[33] M. Belkin and P. Niyogi, "Laplacian eigenmaps and spectral techniques for embedding and clustering," in *Proc. Adv. Neural Inf. Process. Syst.*, 2002, pp. 585–591.

[34] S. Yan, D. Xu, B. Zhang, H.-J. Zhang, Q. Yang, and S. Lin, "Graph embedding and extensions: A general framework for dimensionality reduction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 29, no. 1, pp. 40–51, Jan. 2007.

[35] X. He, S. Yan, Y. Hu, P. Niyogi, and H.-J. Zhang, "Face recognition using Laplacianfaces," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 3, pp. 328–340, Mar. 2005.

[36] S. T. Roweis, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, Dec. 2000.

[37] H. Attouch, J. Bolte, P. Redont, and A. Soubeyran, "Proximal alternating minimization and projection methods for nonconvex problems. An approach based on the Kurdyka–Lojasiewicz inequality," *Math. Operations Res.*, vol. 35, no. 2, pp. 438–457, May 2010.

[38] J. Eckstein and D. P. Bertsekas, "On the Douglas–Rachford splitting method and the proximal point algorithm for maximal monotone operators," *Math. Program.*, vol. 55, nos. 1–3, pp. 293–318, 1992.

[39] S. Boyd, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2010.

[40] G. H. Golub and C. F. Van Loan, *Matrix Computations*, vol. 3. Baltimore, MD, USA: The Johns Hopkins Univ. Press, 2012.

[41] F. Dell'Acqua, P. Gamba, A. Ferrari, J. A. Palmason, J. A. Benediktsson, and K. Árnason, "Exploiting spectral and spatial information in hyperspectral urban data with high resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 4, pp. 322–326, Oct. 2004.

[42] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multiband images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[43] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[44] L. Wald, "Quality of high resolution synthesised images: Is there a simple criterion?" in *Proc. 3rd Conf. Fusion Earth Data, Merging Point Meas., Raster Maps Remotely Sensed Images (SEE/URISCA)*, 2000, pp. 99–103.

[45] K. Zhang, M. Wang, and S. Yang, "Multispectral and hyperspectral image fusion based on group spectral embedding and low-rank factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1363–1371, Mar. 2017.

[46] N. Akhtar, F. Shafait, and A. Mian, "Bayesian sparse representation for hyperspectral image super resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3631–3640.

[47] K. Huang, N. D. Sidiropoulos, and A. P. Liavas, "A flexible and efficient algorithmic framework for constrained matrix and tensor factorization," *IEEE Trans. Signal Process.*, vol. 64, no. 19, pp. 5052–5065, Oct. 2016.



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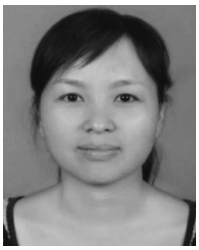
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